Statistics
Parameter Estimation, Hypothesis Testing, Regression

D Manjunath

CMInDS, EE
IIT Bombay

October 5, 2026

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Rules of Engagement

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- Thus statistics is about making inferences from the sample to the population.

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  - To summarise: Statistics are what the investigator **knows** from the samples. Parameters are what they **want** to know.

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Example application: Periodic Labour Force Survey



Wednesday, March 25, 2026 | 05:42 PM IST

Business Standard

Over two-thirds of unemployed Indians aged 20–29 were graduates in 2023, with the share of degree holders among the unemployed rising sharply from 46 per cent in 2017 to 67 per cent in 2023, according to the State of Working India 2026 report released by Azim Premji University o Tuesday.

The report also found that less than half of all graduates were employed, with 48.8 per cent reporting some form of work. Among graduates, only 6.7 per cent had permanent salaried jobs, while 3.7 per cent were in white-collar occupations.

"Between 2004–05 and 2023, while roughly 5 million graduates were added each year, only around 2.8 million found employment, and an even smaller share entered salaried jobs, contributing to rising graduate unemployment and slowing earnings growth," the report observed.

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Example application: Periodic Labour Force Survey

- National Sample Survey Office (NSSO) estimates of employment and unemployment indicators.

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- National Sample Survey Office (NSSO) estimates of employment and unemployment indicators.
- Objective is to determine Labour Force Participation Rate (LFPR), Worker Population Ratio (WPR), and Unemployment Rate (UR).

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  - Convert survey into numbers for India.

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TV Advertising

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- Television rating point (TRP) is used in determining the reach and demographic.

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Example: TV advertising

Hindi TV Serials TRP & FMN Rating

Top 5 TV Serials Of The Week

Week 10, 2026

Rank	Name	TRP	Channel
1	Kyunki Saas Bhi Kabhi Bahu Thi 2	2.0	Star Plus
2	Anupamaa	1.9	Star Plus
3	Tum Se Tum Tak	1.8	Zee TV
4	Taarak Mehta Ka Ooltah Chasmah	1.8	Sab TV
5	Vasudha	1.7	Zee TV

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TRP Ratings

TRP में फर्जीवाड़ा: रिपब्लिक विवाद के बाद ब्रॉडकास्ट काउंसिल BARC ने न्यूज चैनलों की वीकली TRP लिस्ट पर अस्थायी रोक लगाई

मुंबई 6 दिन पहले

CBI Moves To Investigate Fake Ratings Case As Complaint Is Filed In UP

Republic TV has been demanding a CBI probe, accusing the Mumbai police of going after ' after their recent clash over the Sushant Singh Rajput case.

...ported by Arvind Gunasekar, Sreenivasan Jain, Edited by Deepshikha Ghosh | Updated: October 21, 2020 7:07 am IST

TRP scam case: BARC says disappointed with Republic TV for 'disclosing; misrepresenting' confidential communication

The statement comes after the Republic Media Network earlier in the day claimed that BARC, in an email said it had found no complaint or evidence of malpractice against Republic TV, RepublicBharat or any of its affiliates

FP Staff | October 18, 2020 21:23:33 IST



TRP Ratings Scam: CBI Registers Case to Probe Alleged Fraud Based on Complaint Filed in UP

The CBI probe into the case comes amid an ongoing investigation by the Mumbai Police. Three television news channels are under the latter's radar, including the Republic TV headed ...ief Arnab Goswami.



The Manipulation of Television Rating Points: How TRPs work, the scam

Mumbai police are looking at alleged manipulation of Television Rating Points. How do channels score TRPs? What are the ways in which manipulation is possible, and how often has it been alleged in the past?

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TRP ratings: background

- TV adspend was estimated at 27,000 crore rupees in 2020. In 2025, the estimate is more than 40,000 crores

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- There are about 320 (205 rural, 115 urban) million homes in India. Out of this 230 homes have TVs. 97% of these homes are single TV homes,

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- *Reach* and *Target* audience—*Gross Rating Point* and *Target Rating Point* are used by advertisers to buy ad time.

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TRP ratings: organisations involved

- In India, the popularity of different programs are measured with a number called the TRP (Televising Rating Point) rating. It is indicative of TV viewership of a program.

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- Broadcast Audience Research Council (BARC) is a cross industry body that oversees the TV audience measurement. Telecom Regulatory Authority of India (TRAI) regulates the process and provides guidelines for operation.

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TRP ratings: some methodology

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- TRP rating determines the price of advertising in the slot.

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TRP ratings: issues

- An important question for advertiser: Do the same people watch the ad at different times?

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- Effect of chasing TRP rating: Even the definition of news changes—entertainment and crime is more than 30% of the news!

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TRP ratings: some more economics

- It costs more than twenty five crore rupees per year to obtain and analyse the data annually.

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On Sampling

- Estimating the parameters from the sample is justified if the sample represents the population.

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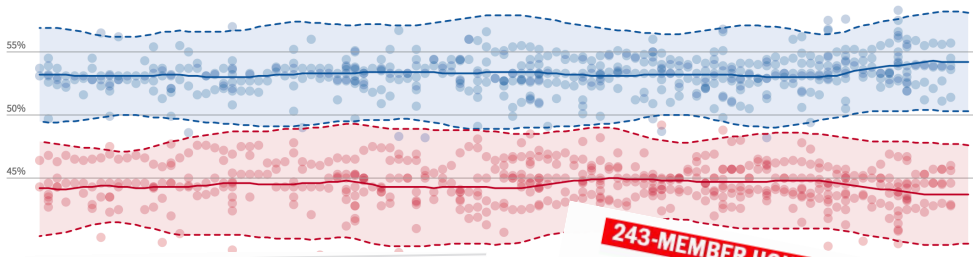
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- And how bad: In the 1948 US Presidential election (Harry Truman vs Thomas Dewey) three separate polls said Dewey would get 50, 50, and 53 % of the vote and Truman would get 45, 44, and 38% of the vote. Actual result: 50% for Truman and 45% for Dewey! And here is the famous picture.

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US presidential election 2020: is Donald Trump or Joe Biden leading in our poll tracker?

Last updated: October 21

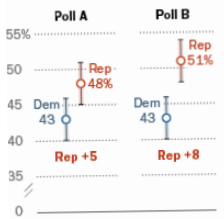
See how the opinion polls have shifted over the course of the election campaign. Each dot represents the result of a published voting intention survey while the solid lines represent our weighted polling average and the dotted lines show margin of error.



For election polls, different measures of the race have different margins of error

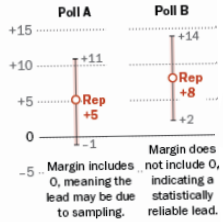
The margin of error reported for most polls applies to support for individual candidates ...

Margin of error for single candidate support (MOE +/– 3 pct. points)



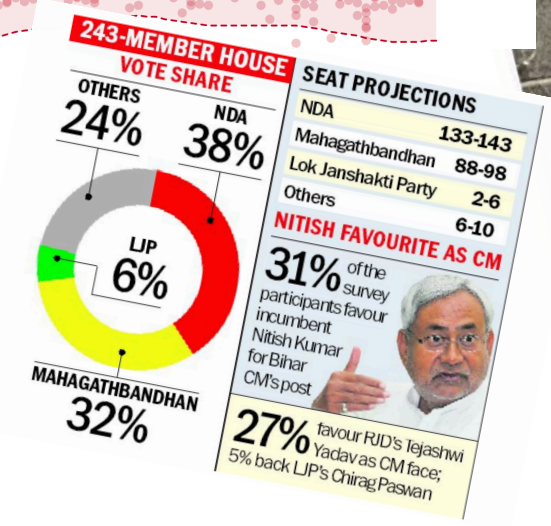
... while the margin of error for a candidate's lead is nearly twice as large.

Margin of error for difference between two candidates' level of support (%Rep – %Dem) (MOE +/– 6 pct. points)



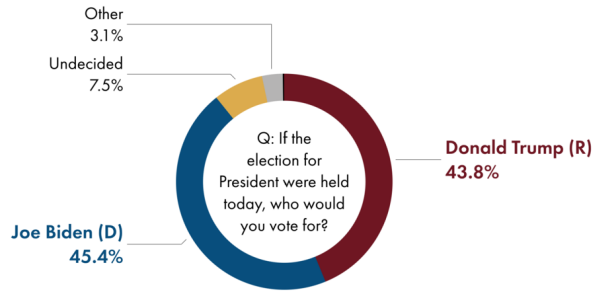
Source: Hypothetical polling results from a fictitious election.

PEW RESEARCH CENTER





Trump vs. Biden spread within the margin of error in Arizona



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A Running Experiment: Impact of MDR on UPI

- Transactions above 2000 in value at big merchants will attract an MDR of 0.4%

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- Think of experiments to check if these things have indeed happened.

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A Running Experiment: Impact of MDR on UPI

- Transactions above 2000 in value at big merchants will attract an MDR of 0.4%
- What 'negative' things can happen? Make a list by listening to criticisms.
- Think of experiments to check if these things have indeed happened.
- Keep refining the questions. We will have a discussion at the end.

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Preview of the Rest of the Course

- Three key topics: Estimation, Hypothesis testing, and Regression.

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- Regression: propose a relation between a set of explanatory variables and 'response variable' determine parameters of the model and/or test hypotheses about the relation.
- Key concepts to be used: pdf and pmf, Independent random variables, moments, CLT and LLN to provide some intuition.

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Recap of Probability Concepts

- Properties of probability measure

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Recap of Probability Concepts

- Expectation and Variance of random variables, and their properties

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- Much of the material is covered in the text *Introduction to Probability and Statistics for Engineers and Scientists,* by Sheldon Ross.

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Gaussian Distribution

- Recall that the probability density function of a Gaussian random variable Z with mean μ and variance σ^2 is

$$f_Z(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$

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- Multivariate Gaussians have a nice structure.

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- Linear transformations of Gaussian random variables results in a Gaussian random variables.
- Uncorrelated Gaussian random variables are independent
- Conditional expectation of a Gaussian is linear

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Law of Large Numbers

- Let X_1 be a random variable with mean (expectation, average) μ and variance σ^2.

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- Let X_1 be a random variable with mean (expectation, average) μ and variance σ^2.
- Let X_i for $i = 1, \ldots, n$ be n samples of the random variable.

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$$\begin{aligned} S_n &= \sum_{i=1}^{n} \mathsf{X}_n \\ M_n &= \frac{S_n}{n} \end{aligned}$$

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- Natural question: What happens to M_n as n becomes large

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- Intuitive answer: It becomes closer to μ.
- What does this mean?

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- As a first approximation this means that the variance of M_n decreases with increasing n.

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- Recall that the variance of $\sum_{i=1}^{n} X_n$ will be $n\sigma^2$ and hence the variance of M_n would be $\frac{1}{n^2}\ n\sigma^2 \ = \ \frac{\sigma^2}{n}$.

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- This is the Law of Large numbers

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Central Limit Theorem

- Let us now consider a slightly different "scaling" and shifting

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- Recall that $\mathsf{E}(S_n - n\mu) = 0$ and

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- As n becomes large s_n 'approaches' a unit normal random variable.

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- As n becomes large s_n 'approaches' a unit normal random variable.
- This means that for 'large-ish' n s_n can be approximated by a unit normal.

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Limit Theorems

- LLN and CLT are called limit theorems.

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- CLT justifes the widespread use of, and interest in, the Gaussian distribution.

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